LEARNER GUIDE

Finding a side

Introduction to Trigonometry. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

Until now you have worked out ratios from triangles you could already see. This lesson turns that around: you use a ratio to find a side you do not know, from just one side and one angle. There is no new rule to memorise — you write the ratio as an equation, put in the numbers with the unknown as x, and use algebra to solve for x. Every solution is the same four lines: (1) write the ratio, e.g. sin θ = opposite/hypotenuse; (2) put in what you know, e.g. sin 40° = x/8; (3) rearrange to get x on its own, e.g. x = 8 × sin 40°; (4) work it out on a calculator (set to degrees). You will use this to reach lengths you could never measure with a ruler — the height of a tree, a cliff, a flagpole. The question you are exploring is: how do I find a length from one side and one angle?

What you are learning to notice

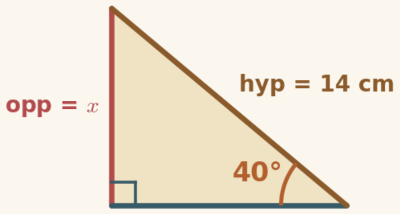
- That one angle and one side are enough to find another side, using the right ratio.
- That you write the ratio as an equation (with the unknown as x) and solve it with algebra — there is no rule to remember.
- That when line 2 is ratio = x ÷ known, you multiply both sides by the known number to get x on its own.
- That the same four lines work every time: write the ratio, substitute, rearrange, evaluate (calculator in degrees).

Moving through it, one step at a time

In Set it up, choose the ratio whose two sides are the known one and the unknown one, then reveal the working one line at a time and watch the algebra unfold. In Side Finder, do it yourself: choose the ratio (which writes lines 1 and 2), then type the rearrangement line that gets x on its own, then work out the answer on a calculator to 1 decimal place — each line stays on screen below the last. In Set up or slip?, judge two lines of someone's working — right ratio, and correct algebra (multiply, not divide)? In Reach the unmeasurable, use the same four lines on real heights you could never measure directly. Keep your calculator in degrees, and use the Scratchpad to write out each solution line by line.

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

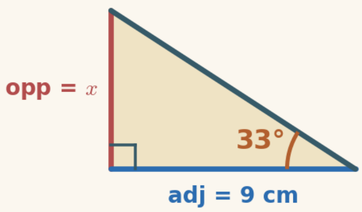


Example 1 — finding a side with sine

You know the angle (40°) and the hypotenuse (14 cm), and want the opposite side. Opposite and hypotenuse → sine.

1. write the ratio $\sin 40° = \dfrac{\text{opposite}}{\text{hypotenuse}}$
2. put in the numbers $\sin 40° = \dfrac{x}{14}$
3. rearrange (x is in the numerator, so multiply) $x = 14 \times \sin 40°$
4. work it out $x = 14 \times 0.643$

x = 9.00 cm



Example 2 — finding a side with tangent

This time you know the angle (33°) and the adjacent side (9 cm), and want the opposite. Opposite and adjacent → tangent.

1. write the ratio $\tan 33° = \dfrac{\text{opposite}}{\text{adjacent}}$
2. put in the numbers $\tan 33° = \dfrac{x}{9}$
3. rearrange (x is in the numerator, so multiply) $x = 9 \times \tan 33°$
4. work it out $x = 9 \times 0.649$

x = 5.84 cm

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (the three ratios and SOH · CAH · TOA (Lesson 4), naming the sides relative to an angle (Lesson 2), reading a trig value from a calculator in degrees, and rearranging a simple equation (multiplying both sides)), open a short recap and return whenever you are ready.

- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

NEO by Nudge Education · nudgeeducation.online · Curriculum by Gerry Docherty. Learner Guide v0.1.